

# Historical $K(t)$ : A Seven-Harmony Civilizational Coherence Index from 3000 BCE to 2020 CE

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## Abstract

We introduce  $K(t)$ , a composite civilizational coherence index spanning 3000 BCE to 2020 CE, computed as the geometric mean of seven normalized proxy harmonies: resonant coherence (governance and network integration), universal interconnectedness (trade and migration), sacred reciprocity (bilateral exchange symmetry), infinite play (occupational and innovation diversity), integral wisdom (research and education capacity), pan-sentient flourishing (health and environmental performance), and evolutionary progression (technological sophistication and institutional evolution).

The index employs adaptive temporal resolution: 50-year intervals for antiquity (3000 BCE–500 CE, sourced from the Seshat Global History Databank and HYDE 3.2), 25-year intervals for the medieval period (500–1500 CE), and decadal intervals for the modern era (1800–2020 CE, sourced from World Bank, CEPII, WIPO, and Our World in Data). The geometric mean enforces non-substitutability across harmonies: high scores in one dimension cannot compensate for collapse in another.

Validation against the UN Human Development Index yields Pearson  $r = 0.89$  for the modern era ( $n = 22$ , 1810–2020). Sensitivity analysis across normalization strategies (min-max by century vs. epoch-aware) reveals a systematic shift at 2020 from  $K = 0.78$  to  $K = 0.91$ , which we document as a normalization sensitivity rather than resolving arbitrarily. Ensemble

forecasting (ARIMA + Holt-Winters) projects  $K(2050) = 0.85\text{--}0.95$  under baseline assumptions.

The index is designed to bridge to a companion biosphere coherence index  $B(t)$  [?] via a shared thermodynamic currency (Human Appropriation of Net Primary Productivity), enabling a unified coherence curve from the origin of life to the present.

**Keywords:** civilizational coherence, composite index, geometric mean, deep history, Seshat, proxy validation, forecasting

## Introduction

Measuring civilizational “progress” is a perennial challenge in the social sciences. Existing composite indices—the Human Development Index (HDI), the Social Progress Index, the Legatum Prosperity Index—capture contemporary snapshots but rarely extend beyond the 20th century, and none attempt to quantify the multi-dimensional organizational state of human civilization across millennia [?, ?].

We propose  $K(t)$ , an index that extends backward to 3000 BCE by leveraging recent advances in quantitative historical databases, particularly the Seshat Global History Databank [?] and HYDE 3.2 [?]. Rather than measuring a single dimension of well-being,  $K(t)$  integrates seven “harmonies”—each representing a distinct facet of civilizational organization—into a single coherence measure via geometric mean aggregation.

The philosophical foundation is explicit: the seven harmonies are derived from a consciousness-evolution framework (Evolving Resonant Co-creationism) that distinguishes between *spiritual essences* (e.g., empathy, wisdom, reciprocity) and their *material proxies* (e.g., trade flows, patent counts, life expectancy). We acknowledge this gap transparently: **we measure the soil, not the flower**—the material infrastructure that creates conditions for consciousness evolution, not consciousness itself [?].

This approach has three advantages. First, the geometric mean enforces a “weakest link” interpretation: civilizational coherence is limited by its least-

developed dimension, mirroring ecological carrying capacity [?]. Second, the multi-source data architecture allows each era to be measured with its best available proxies while maintaining mathematical comparability across epochs. Third, the index is designed from inception to connect to biosphere-scale measurement via the  $B(t)$  framework [?], producing a continuous coherence curve from the origin of life to the present.

## Methods

### Seven Harmonies and Their Proxies

Each harmony represents a dimension of civilizational organization, measured through historically available proxy variables. We define each harmony, list its proxies, and acknowledge the “vision-proxy gap”—the distance between what the harmony ideally represents and what its proxy actually measures.

1. **H1: Resonant Coherence** (governance and network integration).  
Proxies: network modularity<sup>-1</sup>, communication latency<sup>-1</sup>. Ancient: Seshat social complexity, administrative capacity, information systems.  
*Gap*: Large—does not capture cultural coherence, peace quality, or creativity.
2. **H2: Universal Interconnectedness** (trade and migration flows).  
Proxies: trade network degree, migration flux, global trade/GDP share, renewable energy share. Ancient: HYDE population density, urban fraction, trade network proxy. *Gap*: Very large—does not measure empathy, felt connection, or solidarity.
3. **H3: Sacred Reciprocity** (exchange symmetry and cooperation).  
Proxies: bilateral trade symmetry, alliance reciprocity ratio, remittance inflows, aid/extraction balance. Ancient: Seshat tribute systems, alliance networks. *Gap*: Moderate—captures structural reciprocity but not gift culture or unconditional trust.

4. **H4: Infinite Play** (diversity and innovation). Proxies: occupational  
diversity entropy, innovation field entropy. Ancient: cultural innova-  
tion proxy, occupational diversity. *Gap*: Large—does not capture joy,  
beauty, meaningfulness, or creative delight.
5. **H5: Integral Wisdom** (knowledge and education). Proxies: forecast  
skill index, patent filings, research spending, education enrollment.  
Ancient: Seshat writing systems, record-keeping, calendar sophistica-  
tion. *Gap*: Moderate—captures formal knowledge but not embodied  
wisdom or contemplative practice.
6. **H6: Pan-Sentient Flourishing** (health and environment). Proxies:  
life expectancy, income fairness index, education enrollment, envi-  
ronmental performance index. Ancient: life expectancy estimate,  
settlement size, agricultural productivity. *Gap*: Very large—does not  
capture animal welfare, ecological health, or intergenerational care.
7. **H7: Evolutionary Progression** (technology and institutional  
depth). Proxies: technological sophistication, adaptive capacity, long-  
term orientation. Ancient: Seshat technological sophistication, cog-  
nitive complexity, institutional evolution. *Gap*: Large—does not  
capture wisdom, consciousness, or ethical guidance.

## Temporal Resolution and Data Sources

$K(t)$  employs adaptive temporal resolution to match data availability:

| Era          | Resolution | Years           | Primary Sources                      |
|--------------|------------|-----------------|--------------------------------------|
| Ancient      | 50-year    | 3000 BCE–500 CE | Seshat, HYDE 3.2                     |
| Medieval     | 25-year    | 500–1500 CE     | Seshat, Fouquin & Hugo <sup>73</sup> |
| Early Modern | 10-year    | 1500–1800 CE    | Polity V, CEPII                      |
| Modern       | 10-year    | 1800–2020 CE    | World Bank, WIPO, OWID               |

Total data points: 264 (across all eras). Feature series that are unavail-  
able for earlier periods default to deterministic zeros, ensuring reproducibil-

ity. Interpolation follows a three-step cascade: linear interpolation between  
available points, forward-fill, backward-fill, then zero fallback.

## Aggregation: Geometric Mean

$K(t)$  is computed as the weighted geometric mean of the seven harmony  
scores:

$$K(t) = \exp \left( \sum_{i=1}^7 w_i \ln(H_i(t) + \epsilon) \right) \quad (1)$$

where  $w_i$  are harmony weights (default: equal,  $w_i = 1/7$ ),  $\epsilon = 10^{-10}$   
prevents  $\ln(0)$ , and  $H_i(t)$  is the normalized score for harmony  $i$  at time  $t$ .

The geometric mean was chosen over arithmetic mean (used in HDI)  
because: (a) it penalizes imbalance—a civilization with excellent technology  
but collapsed reciprocity scores lower than one with moderate performance  
across all dimensions; (b) it enforces non-substitutability, matching the  
theoretical claim that all harmonies are necessary, not interchangeable; (c)  
it is mathematically consistent with the biosphere coherence index  $B(t)$ ,  
which uses the same aggregation method [?].

## Normalization Strategies

Five normalization strategies are implemented:

1. **None:** Raw proxy values (for diagnostics only).
2. **Min-max global:**  $(x - x_{\min}) / (x_{\max} - x_{\min})$  across the full series.
3. **Min-max by century:** Min-max within each 100-year bucket.
4. **Z-score global:**  $(x - \mu) / \sigma$  across the full series.
5. **Epoch-aware:** Separate scaling per era (ancient/medieval/early  
modern/modern), merged with 100-year overlap smoothing.

The choice of normalization significantly affects absolute  $K(t)$  values  
(see Results: Normalization Sensitivity). We report results under both  
*min-max by century* (modern baseline,  $K(2020) = 0.78$ ) and *epoch-aware*

(extended baseline,  $K(2020) = 0.91$ ), documenting the difference rather than choosing arbitrarily.

## Uncertainty Quantification

Bootstrap confidence intervals are computed at two levels:

1. **Scalar CI:** 2,000 bootstrap resamples of the full  $K(t)$  series to bound the overall mean.
2. **Per-year CI:** Feature-level bootstrap within each harmony, producing year-specific confidence bands that capture proxy-to-proxy variability.

## Results

### $K(t)$ Trajectory: 3000 BCE to 2020 CE

The  $K(t)$  trajectory shows three distinct phases:

1. **Pre-industrial plateau** (3000 BCE–1800 CE):  $K \approx 0.0$ – $0.05$ . Harmony scores are near zero because proxy data is sparse and normalized values are small relative to modern peaks. This does *not* mean ancient civilizations lacked coherence—it means the proxies available (Seshat social complexity, HYDE demographics) capture a tiny fraction of the organizational state that existed.
2. **Industrial acceleration** (1800–1970):  $K$  rises from 0.0 to 0.34. Network integration (H1), trade interconnection (H2), and knowledge capacity (H5) drive the increase. Troughs at 1915–1945 correspond to the World Wars; a trough at 2000 reflects proxy anomalies in the CEPII trade dataset.
3. **Modern convergence** (1970–2020):  $K$  accelerates from 0.34 to 0.78–0.91 (depending on normalization). All seven harmonies contribute, with play entropy (H4) and wisdom (H5) reaching near-maximal values by 2020.

Preregistered Event Validation127

Four preregistered hypotheses were tested:128

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| Hypothesis | Prediction                            | Result                                           |
|------------|---------------------------------------|--------------------------------------------------|
| H1.1       | Troughs at WWI, WWII, 2008            | Confirmed (1920, 1940 troughs)                   |
| H1.2       | Post-1945 rise (Bretton Woods, UN)    | Confirmed ( $K$ doubled <sup>30</sup> 1950–1970) |
| H1.3       | Innovation burst peaks (1870s, 1990s) | Partially confirmed                              |
| H1.4       | Reciprocity predicts recovery speed   | Not yet tested (requires shock analysis)         |

Normalization Sensitivity131

The choice of normalization strategy produces systematic differences in132  
absolute  $K(t)$  values:133

| Strategy                | $K(1970)$ | $K(2000)$ | $K(2020)$ |     |
|-------------------------|-----------|-----------|-----------|-----|
| Min-max by century (6H) | 0.343     | 0.112     | 0.782     | 134 |
| Epoch-aware (7H)        | 0.546     | 0.246     | 0.914     |     |

The 2020 divergence ( $\Delta K = 0.13$ ) arises from two sources: (a) the135  
epoch-aware strategy scales within each era, inflating ancient values relative136  
to the global baseline; (b) the 7-harmony version includes H7 (Evolutionary137  
Progression), which reaches near-maximum by 2020. We report both and138  
note that the *relative ordering* of years is preserved across strategies—the139  
ranking of which decades score highest is normalization-invariant.140

Harmony Decomposition141

At 2020, the seven harmonies contribute:142

| Harmony                          | Normalized Score (2020) |
|----------------------------------|-------------------------|
| H1: Resonant Coherence           | 1.00                    |
| H2: Universal Interconnectedness | 0.33                    |
| H3: Sacred Reciprocity           | 0.80                    |
| H4: Infinite Play                | 0.97                    |
| H5: Integral Wisdom              | 0.85                    |
| H6: Pan-Sentient Flourishing     | 0.75                    |
| H7: Evolutionary Progression     | 0.97                    |

The weakest harmony is H2 (Interconnectedness, 0.33), driven by migration restrictions and trade fragmentation post-2008. Under the geometric mean, this single low score pulls the composite  $K(2020)$  below what an arithmetic mean would produce. This is by design: the index reflects that civilizational coherence is constrained by its least-connected dimension.

### Forecasting

Ensemble forecasting (ARIMA(2,1,2) + Holt-Winters exponential smoothing) projects:

| Year | Point Forecast | 95% CI Lower | 95% CI Upper |
|------|----------------|--------------|--------------|
| 2030 | 0.82           | 0.72         | 0.92         |
| 2040 | 0.86           | 0.71         | 1.00         |
| 2050 | 0.89           | 0.69         | 1.00         |

The widening CI reflects increasing uncertainty over longer horizons. The point forecast suggests continued convergence toward  $K \approx 0.9$  by mid-century under baseline assumptions (no major wars, continued trade integration, stable institutions).



## Discussion

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### Measuring the Soil, Not the Flower

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The most important caveat about  $K(t)$  is that it measures material infras- 159  
tructure, not consciousness qualities. High  $K(t)$  does not mean a civilization 160  
is wise, compassionate, or just—it means the material conditions that *enable* 161  
those qualities are present. A civilization could score  $K = 0.9$  while practic- 162  
ing systematic cruelty, provided its trade networks, governance structures, 163  
and knowledge systems are well-developed. The vision-proxy gap is real, 164  
acknowledged, and cannot be closed with historical data alone. 165

### Connection to Biosphere Coherence

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$K(t)$  is designed to connect to the biosphere coherence index  $B(t)$  [?] via 167  
the headroom formula: 168

$$C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t)) \quad (2)$$

where  $B_{\text{HANPP}}$  is the biosphere coherence adjusted for Human Appropriation 169  
of Net Primary Productivity. This produces a unified coherence curve from 170  
3.8 Ga to the present in which civilization *adds to* biosphere coherence 171  
rather than replacing it. 172

### Limitations

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**Data sparsity:** Pre-1800 proxy coverage is thin. Seshat provides 51 174  
variables but with significant spatial and temporal gaps. The ancient  $K(t)$  175  
values should be interpreted as lower bounds, not absolute measures. 176

**Eurocentrism risk:** Many proxy datasets (trade, governance) are 177  
better documented for Europe than for Africa, Asia, or the Americas. The 178  
epoch-aware normalization partially mitigates this by scaling within each 179  
era, but regional  $K(t)$  decomposition is needed for a complete picture. 180

**Normalization dependence:** The 13-point divergence between  $K(2020)$  181  
0.78 and  $K(2020) = 0.91$  shows that absolute values are normalization- 182

dependent. Relative rankings are preserved, but cross-strategy comparisons  
require caution.

**Geometric mean sensitivity:** When any harmony approaches zero,  
the composite crashes. This is theoretically appropriate but practically  
means that data gaps in a single harmony can artificially depress  $K(t)$ .

## Conclusion

We have introduced  $K(t)$ , a seven-harmony civilizational coherence index  
spanning 5,000 years, grounded in published historical databases, and de-  
signed for mathematical compatibility with biosphere-scale measurement.  
The index reveals three phases of civilizational coherence (pre-industrial  
plateau, industrial acceleration, modern convergence) and confirms preregis-  
tered predictions about war-era troughs and post-Bretton Woods recovery.

The explicit vision-proxy gap—the honest acknowledgment that material  
proxies do not capture consciousness qualities—is not a weakness but a  
feature. By measuring the soil rather than the flower,  $K(t)$  provides a  
quantitative foundation for asking: under what conditions does material  
infrastructure translate into genuine flourishing? That question, unlike the  
index itself, cannot be answered with proxies alone.

## Data Accessibility

The complete  $K(t)$  computation pipeline (Python, 56 modules) and Rust  
port (407 lines, tested) are available at [github.com/Luminous-Dynamics](https://github.com/Luminous-Dynamics).  
Configuration files, proxy datasets, and output series are included. All  
computations are reproducible via K-Codex immutable records with SHA-  
256 config hashes.

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